Lab 4: Second-Order Systems, and Unity Feedback, PD and PV Controllers

Objectives

You study the effects of numerical differentiation on clean and noisy signals, and observe how differentiation amplifies noise and sudden changes in a signal. This analysis is important because the derivative term in a PID controller requires taking the derivative of a signal.

You study a second-order system with unity feedback and with PD and PV controllers, and measure the peak time and percent overshoot of the response using both the DC motor model and the physical DC motor. You design and implement PD and PV controllers with a 1st order filter, and you relate the effects of k_p and k_d to the natural frequency ω_n and damping ratio ζ.

Finally, you design k_p and k_d to meet a stated specification on overshoot, peak time and motor voltage, and you keep revising the design on the hardware until it meets that specification or you can explain why it cannot.

From this lab, you learn that the natural frequency ω_n and damping ratio ζ are key characteristics of a system's response. With unity feedback, these parameters are fixed by the system. However, by implementing PD or PV controllers, you can design suitable k_p and k_d values to achieve desired ω_n and ζ, thereby shaping the system response.

Timeline

This lab spans two weeks. Complete the pre-lab tasks **before the first lab session**. Review all reading materials **during the two weeks but before attending each lab session**. Do not wait until the lab sessions to complete these preparations.

Bring with you

- **Your Lab 3 results** — the motor resistance R_m and back-emf constant k_m you identified. Section 7 asks you to recompute the model gain K and time constant τ from them and to compare the result with the data-sheet values.
- **Your Lab 2 Simulink models.** Section 6.3 starts from the filtering model you built in Lab 2; you are not given a model to open.

Pre-Lab Tasks

Due: Before the first lab session. Note: You are not required to upload anything to D2L. This pre-lab is for your own practice and preparation only.

Complete the following tasks:

- MATLAB course to take, PID Control Techniques, about 30 minutes
  `https://matlabacademy.mathworks.com/details/pid-control-techniques/otmlslpct`
- Using `PIDGainEffects.mlx`, posted to D2L, practice MATLAB programming on PID, step response, closed-loop feedback control, and the effects of control action gains.

Work with this code to observe the effects of the gains *before* you spend bench time changing them on the hardware.

- Learn how to use the MATLAB `diff` function.
  `https://www.mathworks.com/help/matlab/ref/double.diff.html`
- Learn how to use the MATLAB `stepinfo` function. You will use it to report peak time, percent overshoot and settling time in Sections 6.2 and 7, rather than reading values off a scope by eye.
  `https://www.mathworks.com/help/control/ref/dynamicsystem.stepinfo.html`
- **Be able to move in both directions** between a time-domain specification and the second-order parameters: given a required percent overshoot and peak time, find ζ and ω_n; and given ζ and ω_n, find k_p and k_d. Section 7 is a design-to-specification task, and you will not have time to work this out for the first time at the bench.

Required Reading

Complete during the two-week lab period, before attending lab sessions.

Reference book "Web Appendices", accessible via `www.fpe8e.com`: Appendix WA, A Review of Complex Variables (*Z01 FRAN5717 08 SE WA.pdf*), including

- WA.1 Definition of a Complex Number,
- WA.2 Algebraic Manipulations,
- WA.3 Graphical Evaluation of Magnitude and Phase

From the reference book, *Feedback Control of Dynamic Systems* (8th edition), review the following topics:

1. Second-order systems (damping ratio, natural frequency, peak time and overshoot): Effects of Pole Locations (Section 3.3) and Time-Domain Specifications (Section 3.4).
2. Second-order systems (steady-state gain, time constant).
3. PID control, and open- and closed-loop systems: The Basic Equations of Control (Section 4.1, pages 188–189), The Three-Term Controller (Sections 4.3.1 to 4.3.5). Within Section 4.3, pay particular attention to the derivative term: why differentiating a measured signal is a problem in practice, why the derivative is filtered, and why feeding back the derivative of the *output* rather than of the *error* avoids a derivative kick. Section 6.3 of the lab assumes you have read this.

What the lab will ask you to design

Sections 6.1 to 6.3 build the tools. **Section 7 is where the design happens**, and it is the largest single part of the mark. You will be given a specification, a maximum percent overshoot, a maximum peak time, and a motor voltage that must not saturate, and asked to:

1. state the requirements and constraints before you build anything,
2. predict the gains that should meet the specification,
3. test them on the hardware, measure what actually happens, and revise,
4. record every iteration, including the ones that failed, with a one-line reason for each change,
5. and explain the trade-offs you had to balance.

Your first design will almost certainly not meet the specification. That is expected. What is graded is the reasoning behind each revision, not how quickly you converge. A design that misses the specification and diagnoses the reason correctly is worth more than one that meets it with no explanation.